

Companion LXXVI: Wasserstein Information Geometry of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Wasserstein geometry provides the natural metric structure on the space of persistence diagrams, capturing the optimal transport cost required to match topological features across diagrams. Information geometry, in contrast, equips statistical manifolds with a Riemannian structure derived from Fisher information. This Companion develops a unified Wasserstein information geometry for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, altering both the Wasserstein transport structure and the information-theoretic curvature of topological ensembles. The resulting framework provides a geometrically rich and observationally relevant description of RDCC signatures in weak-lensing topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Wasserstein distances provide the canonical metric on the space of diagrams, measuring the optimal transport cost required to match topological features across realizations. Previous Companions developed Hilbert-space embeddings, kernel PCA, MMD, and Fisher information geometry as tools for analyzing the Relaxation-Driven Cyclic Cosmology (RDCC).

Wasserstein information geometry synthesizes these perspectives by equipping the Wasserstein space of persistence diagrams with an information-theoretic Riemannian structure. This structure quantifies how sensitively the optimal transport geometry responds to changes in cosmological parameters. In RDCC, the relaxation mechanism modifies the gravitational potential in a scale-dependent manner, producing characteristic signatures in the Wasserstein information geometry of topological ensembles.

2 Wasserstein Geometry of Persistence Diagrams

Given two persistence diagrams D and E , the p -Wasserstein distance is

$$W_p(D, E) = \left(\inf_{\gamma} \sum_{x \in D} \|x - \gamma(x)\|^p \right)^{1/p}, \quad (1)$$

where γ ranges over bijections between D and E (including the diagonal).

This metric captures the minimal transport cost required to match topological features across diagrams.

3 Information Geometry on Wasserstein Space

Let $p(D|\theta)$ be a statistical model for persistence diagrams. The Fisher information matrix is

$$I_{ab}(\theta) = \mathbb{E} [\partial_{\theta_a} \log p(D|\theta) \partial_{\theta_b} \log p(D|\theta)]. \quad (2)$$

The Wasserstein information metric is defined by pulling back the Fisher metric through the Wasserstein map:

$$g_{ab}^W(\theta) = \mathbb{E} [\partial_{\theta_a} W_p(D, \mu(\theta)) \partial_{\theta_b} W_p(D, \mu(\theta))], \quad (3)$$

where $\mu(\theta)$ is the Wasserstein barycenter of the distribution.

4 RDCC Modification of Wasserstein Structure

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

This modifies the birth and death coordinates of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b(\ell/\ell_{\text{rel}})^\alpha], \quad (5)$$

$$d_{\text{RDCC}} = d [1 - \chi_d(\ell/\ell_{\text{rel}})^\alpha]. \quad (6)$$

The Wasserstein distance becomes

$$W_p^{\text{RDCC}}(D, E) = W_p(D, E) \left[1 - \chi_W \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

5 Wasserstein Information Geometry in RDCC

The RDCC Wasserstein information metric is

$$g_{ab}^{W, \text{RDCC}} = g_{ab}^W \left[1 - \chi_{g, ab} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (8)$$

RDCC induces:

- reduced Wasserstein curvature at small scales,
- enhanced curvature at large scales,
- a characteristic turnover near ℓ_{rel} ,
- anisotropic deformation of Wasserstein geodesics,
- modified barycentric flow of topological ensembles.

6 Conclusion

Wasserstein information geometry provides a powerful framework for analyzing the optimal-transport sensitivity of persistence diagrams in the Relaxation-Driven Cyclic Cosmology. The relaxation mechanism modifies both the Wasserstein transport structure and the information-theoretic curvature of topological ensembles in a scale-dependent manner. These effects offer a geometrically rich and observationally relevant probe of the RDCC gravitational field. The present Companion establishes the foundation for future studies of Wasserstein geometry, information geometry, and optimal transport in cosmological topology.

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